

Jack Thomas Parley

Physicist/AI researcher | Ph.D. in Statistical Physics | Advanced Japanese + 6 European Languages

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Profile

AI researcher with a Ph.D. in statistical physics, now working on how deep networks learn language-like structure from examples alone. My research combines probabilistic modeling, grammar induction, sample-complexity theory, and scalable PyTorch implementations of inference algorithms and neural architectures. Current work accepted as a regular paper at ICML 2026, one of the top 3 Machine Learning conferences. I also bring a strongly international profile, including advanced Japanese language ability (JLPT N1) and a multilingual European background.

Core Profile

Research focus: Deep learning, language acquisition, grammar induction, probabilistic modeling, statistical physics of learning.

Technical strengths: PyTorch, Python, scalable scientific computing, probabilistic inference, transformers, inside–outside algorithms.

International profile: Born in Tokyo; native English; fluent Spanish and Catalan; advanced Japanese (JLPT N1), German, and Italian; upper-intermediate French; studied and worked across Spain, France, Italy, Germany, and Switzerland.

Professional Experience

Postdoctoral Researcher, AI / Deep Learning, EPFL, Lausanne, Oct. 2024–present
Switzerland.

- Understand how deep networks learn language-like structure from data alone, using synthetic context-free grammars with controlled vocabulary size, grammar depth, branching structure, and ambiguity.
- Developed fully parallelized PyTorch implementations of inside and inside–outside parsing algorithms for large-scale experiments on grammar learning and latent structure recovery.
- Introduced an *Inside Neural Network*, a neural architecture inspired by dynamic-programming inference, achieving up to an order-of-magnitude improvement in sample efficiency over a standard transformer on controlled classification tasks.
- First-author regular paper accepted at the 43rd International Conference on Machine Learning (ICML 2026): “Deep networks learn to parse uniform-depth context-free languages from local statistics” (currently extending theory of learning to GPT-2-style next-token prediction).

Postdoctoral Researcher, Statistical Physics, Institut für Theoretische Physik, University of Göttingen, Göttingen, Germany. Oct. 2022–Sept. 2024

- Developed mean-field theories for ductile and brittle yielding transitions in amorphous solids under shear.
- Built analytical and numerical models of avalanche behavior beyond standard random-field-Ising-model paradigms for driven disordered systems.
- Published first-author work in *Physical Review E* and developed related projects on fatigue failure and critical dynamics.

Selected Technical Skills

Machine learning: PyTorch; neural network architecture design; transformers; GPT-style next-token

prediction; model training and evaluation; synthetic data generation.

Probabilistic modeling: Context-free grammars; PCFGs; inside and inside–outside algorithms; latent-variable inference; sample complexity; statistical learning theory.

Scientific computing: Python, Julia, Fortran, Matlab, Mathematica; GPU/HPC workflows; vectorized tensor algorithms; numerical simulations.

Statistical physics: Disordered systems; mean-field theory; glassy dynamics; avalanches; non-equilibrium statistical mechanics.

Communication: First-author publications in ICML, PRL, PRE, PRM and Phys. Fluids; international talks; university teaching in English and German.

Education

Ph.D. Physics, *summa cum laude*, University of Göttingen, Germany. Oct. 2018–May 2022

Thesis: *Mean field theories for the aging and yielding of athermal amorphous solids*. Developed analytical and numerical theories for slow relaxation, aging, and yielding in amorphous solids. Resulted in first-author publications in *Physical Review Letters*, *Physical Review Materials*, and *Physics of Fluids*. Supervised by Prof. Peter Sollich.

Research Intern, Institut de Physique Théorique, CEA, Saclay, France. Feb. 2018–July 2018

Research internship: mean-field study of spin-glass models for amorphous solids near the yielding transition, using replica methods and numerical simulations (supervised by Dr. Pierfrancesco Urbani).

International Master in Physics of Complex Systems, Paris / Trieste / Torino. 2016–2018

Two-year French–Italian program in complex systems, statistical physics, probability, inference, stochastic processes, optimization, and computational science. Global average: 16.7/20; rank: 3/40.

BSc in Engineering Physics, Universitat Politècnica de Catalunya, Barcelona, Spain. 2012–2016

Thesis on the effect of complex network topology on collective motion. Work published in *Physical Review Letters* **120**, 068303.

Selected Publications

- Jack T. Parley, Francesco Cagnetta and Matthieu Wyart, “Deep networks learn to parse uniform-depth context-free languages from local statistics”, *Proceedings of the 43rd International Conference on Machine Learning*, PMLR 306 (2026).
- Jack T. Parley and Peter Sollich, “Ductile and brittle yielding of athermal amorphous solids: A mean-field paradigm beyond the random-field Ising model”, *Physical Review E* **110**, 045002 (2024).
- Jack T. Parley, Rituparno Mandal and Peter Sollich, “Mean field description of aging linear response in athermal amorphous solids”, *Physical Review Materials* **6**, 065601 (2022).
- Jack T. Parley, Srikanth Sastry and Peter Sollich, “Mean-Field Theory of Yielding under Oscillatory Shear”, *Physical Review Letters* **128**, 198001 (2022).
- Jack T. Parley, Suzanne M. Fielding and Peter Sollich, “Aging in a mean field elastoplastic model of amorphous solids”, *Physics of Fluids* **32**, 127104 (2020).
- M. Carmen Miguel, Jack T. Parley and Romualdo Pastor-Satorras, “Effects of Heterogeneous Social Interactions on Flocking Dynamics”, *Physical Review Letters* **120**, 068303 (2018).

Selected Conferences and Talks

- 43rd International Conference on Machine Learning (ICML), Seoul, South Korea, July 2026. Poster: “Deep networks learn to parse uniform-depth context-free languages from local statistics”.

- Simons Collaboration for the Physics of Learning and Neural Computation, Stanford University, United States, November 2025. Flash talk.
- Avalanche Conference 2024, Espoo, Finland. Oral contribution: “Mean-field mechanical noise approach to plasticity in amorphous solids can capture essential long-range physics”.
- CECAM Flagship Workshop, Lausanne, Switzerland, May 2023. Oral contribution: “Mean field theory of the brittle yielding transition under uniform shear”.
- STATPHYS satellite workshop, Bariloche, Argentina, July 2019. Oral contribution on time-dependent rheology of amorphous materials.
- STATPHYS 27, Buenos Aires, Argentina, July 2019. Oral contribution on amorphous materials with broad mechanical noise.

Languages and International Experience

English: Native.

Spanish / Catalan: Fluent; long-term academic and personal experience in Spain and Catalonia.

Japanese: Advanced proficiency; JLPT N1. Strong cultural familiarity and interest in Japan- and Asia-facing AI research or industry environments.

German: Advanced / C1; professional research and teaching experience in Germany.

Italian: Advanced; completed part of the International Master in Physics of Complex Systems in Italy.

French: Upper-intermediate / B2; academic and professional experience in French-speaking environments.

International background: Born in Tokyo; studied and worked across Spain, France, Italy, Germany, and Switzerland, with extensive experience communicating across cultures in international research environments.

Teaching Experience

- Teaching assistant / exercise-group instructor for courses in complex systems, statistical machine learning, analytical mechanics, thermodynamics and statistical mechanics, advanced statistical physics, and quantum mechanics at EPFL and the University of Göttingen.
- Private tutor in mathematics and English, Barcelona, 2013–2016.

Awards

- Ph.D. Dissertation Prize, Dr. Berliner–Dr. Ungewitter Stiftung foundation, 2022/23.
- Université Paris-Saclay international master’s scholarship, 2017–2018.
- Catalan national award for outstanding grade in university entrance exams, 2012.